PRACTICE SHEET

1. If the latus rectum of an ellipse is equal to one half its minor axis, what is the eccentricity of the ellipse?

(a) $\frac{1}{2}$ (b) $\frac{\sqrt{3}}{2}$

(c) $\frac{3}{4}$ (d) $\frac{\sqrt{15}}{4}$

2. P(2,2) is a point on the parabola $y^2 = 2x$ and A is its vertex. Q is another point on the parabola such that PQ is perpendicular to AP. What is the length of PQ?

(a) $\sqrt{2}$ (b) $2\sqrt{2}$

(c) $4\sqrt{2}$ (d) $6\sqrt{2}$

3. The focal distance of a point on the parabola $y^2 = 12$ x is 4. What is the abscissa of the point?

(a) 1 (b) -1

(c) $2\sqrt{3}$ (d) -2

4. If (2, 0) is the vertex and the y – axis is the directrix of a parabola, then where is its focus?

(a) (0, 0) (b) (–2, 0)

(c) (4,0) (d) (–4, 0)

5. Which one of the following points lies outside the ellipse $(x^2 / a^2) + (y^2 / b^2) = 1$?

(a) (a, 0) (b) (0, b)

(c) (–a,0) (d) (a, b)

6. What is the equation of the parabola, whose vertex and focus are on the x–axis at distance a and b from the origin

11. Consider the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (b > a). Then, which one of the following is correct?

(a) Real foci do not exist

(b) Foci are (± ae, 0)

(c) Foci are (± be, 0)

(d) Foci are (0, ± be)

12. Consider the parabolas $S_1 \equiv y^2 - 4ax = 0$ and $S_2 \equiv y^2 - 4bx = 0$. S_2 will contains S_1, if

(a) a > b > 0

(b) b > a > 0

(c) a > 0, b < 0 but |b|> a

(d) a< 0, b > 0 but b > |a|

13. Equation of the hyperbola with eccentricity 3/2 and foci at (±2,0) is $5x^2 - 4y^2 = k^2$. What is the value of k?

(a) 4 / 3 (b) 3 / 4

(c) $(4/3\sqrt{5})$ (d) $(3/4\sqrt{5})$

14. What is the eccentricity of an ellipse, if its latusrectum is equal to one – half of its minor axis?

(a) 1/4 (b) 1 / 2

(c) $\sqrt{3}$ / 4 (d) $\sqrt{3}$ / 2

15. What does an equation of the first degree containing one arbitrary parameter passing through a fixed point represent?

(a) Circle (b) Straight Line

(c) Parabola (d) Ellipse